

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

PHYSICS

LECTURE

03

Light – Part - 02



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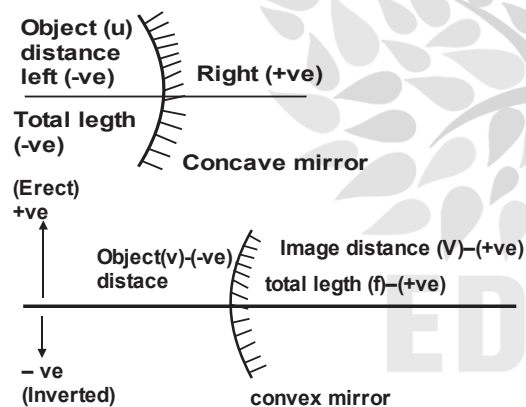
Mirror Formula:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

- f (Focal Length)** → Distance between pole (P) and focus (F)
- v (Image Distance)** → Distance of image from mirror
- u (Object Distance)** → Distance of object from mirror
- Image distance for all cases except one V(-ve)

Sign convention (New Cartesian System)

- All distances measured from pole
- Left side: negative, right: positive
- Heights above principal axis: positive
- Heights below principal axis → Negative



$$\text{Magnifications (m)} = \frac{-v}{u} = \frac{H_i}{H_o}$$

$m > 1$ → Enlarged image

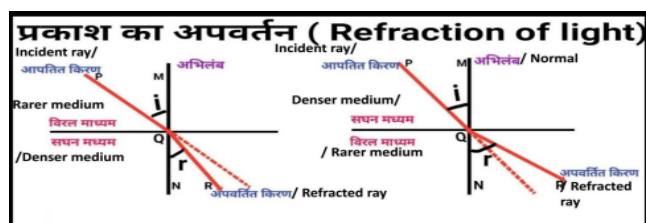
$m < 1$ → Diminished image

m positive → Virtual, erect image

m negative → Real, inverted image

Refraction of Light:

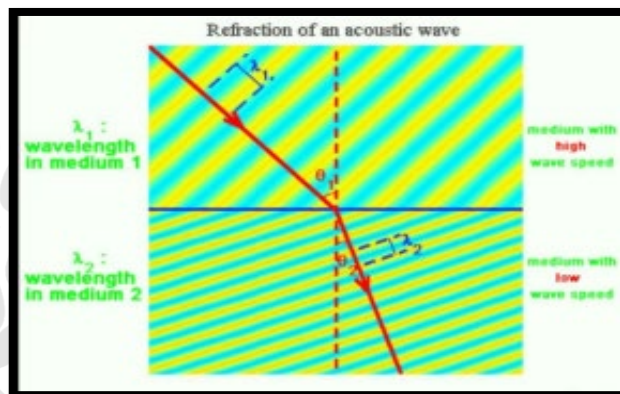
It is the process of Bending of light when it passes from one medium to another.



Laws of Refraction (Snell's Law)

The ratio of the sine of the angle of incidence to the sine of the angle of refraction is constant for a given pair of media.

$$\frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = \mu$$



Refractive Index

$$\mu = \frac{\text{speed of light in vacuum}}{\text{speed of light in medium}}$$

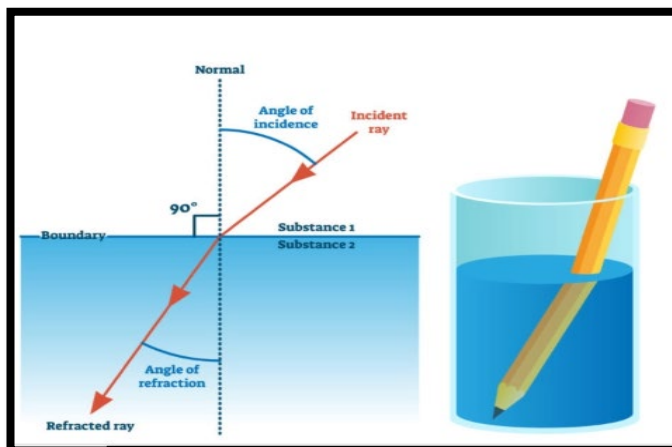
- Light bends towards normal when moving to denser medium
- Light bends away from normal when moving to rarer medium

Example of Refraction of Light:

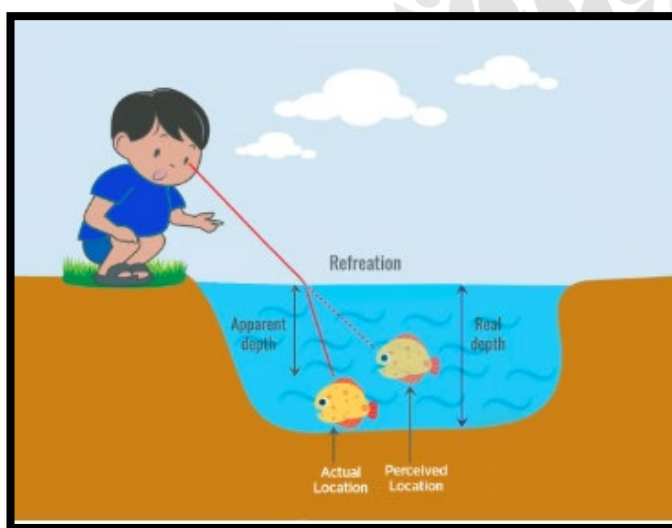
1. Bending of an object

- When pencil in a denser medium is seen from a rarer medium it appears to be bent.

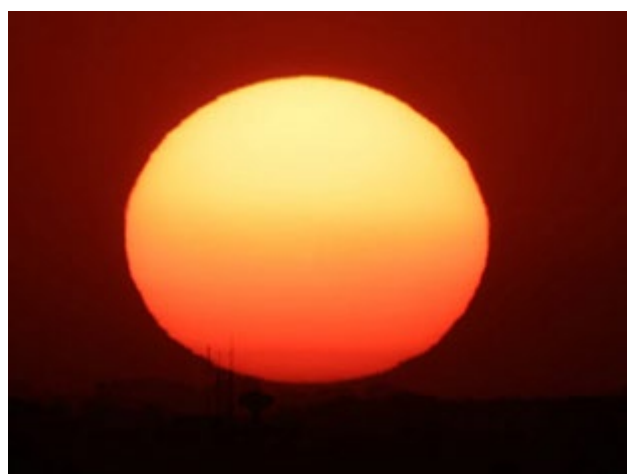




3. Fish appears raised in Water



4. Early sunrise and late sunset



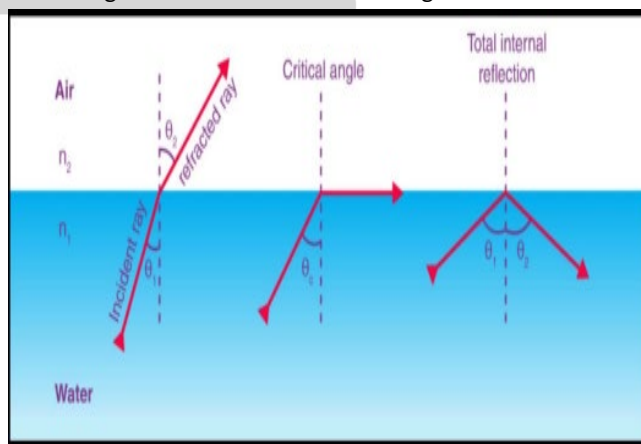
6. Twinkling of Stars

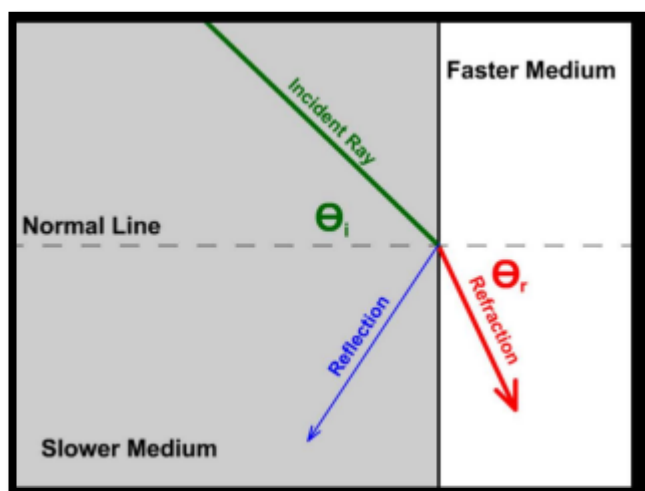
Due to fluctuations in the refractive index of different layers of atmosphere, the refraction becomes irregular so phenomena of twinkling stars occurs.



Total internal reflection

- Total internal reflection is the phenomenon where light traveling from a denser medium to a rarer medium is completely reflected into the denser medium, instead of refracting into the rarer one.
- This occurs when the angle of incidence is greater than the critical angle.

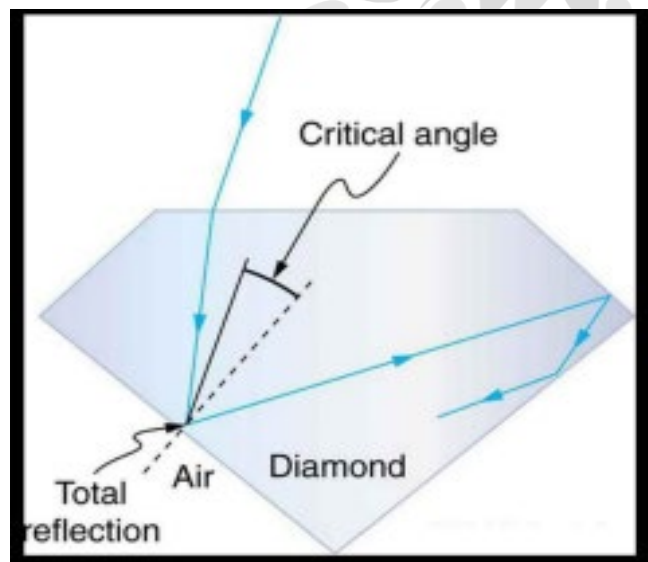




Example of Total Internal Reflection :

I. Sparkling of diamond :

The sparkling of diamond is due to total internal reflection inside it. As refractive index for diamond is 2.5 so θ_c (critical angle) = 24° . Diamond is cut in such a manner that, once the light enters into it, when it tends to come out then $i > \theta_c$ (critical angle). So TIR will take place repeatedly inside it. The light, which beams out from a few places entering into the eyes of the observer makes it sparkle.



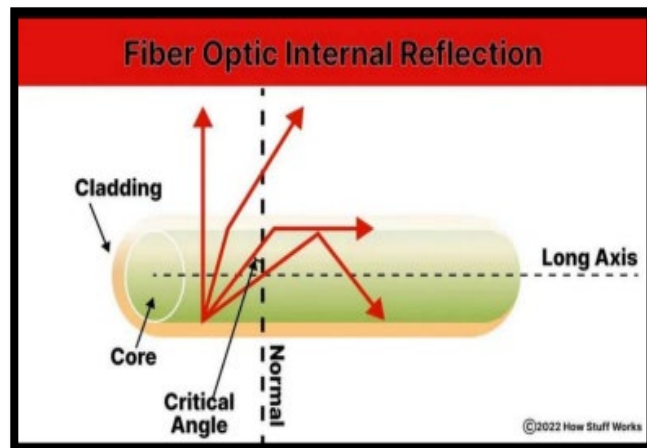
II. Optical Fiber :

In optical fibre light propagates through

m_1, m_2

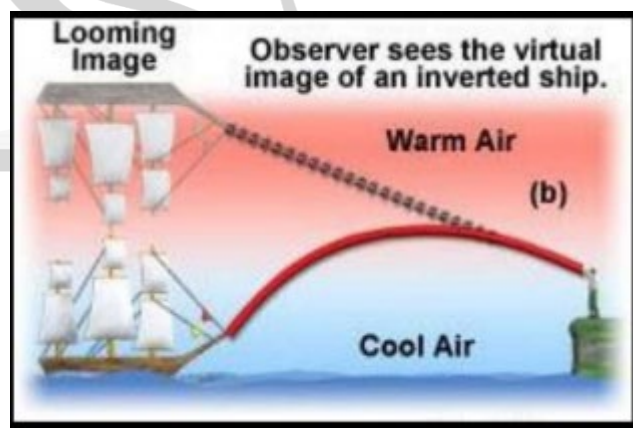
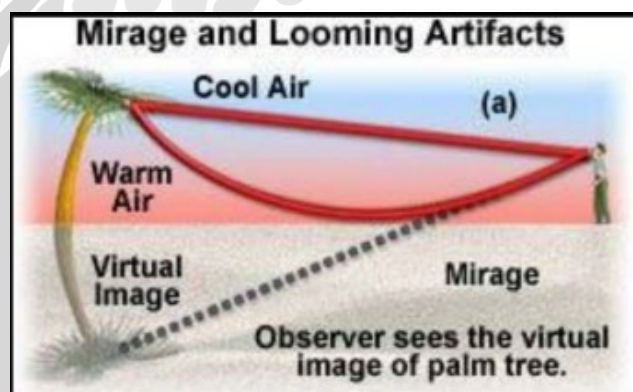
$m_1 > m_2$

light pipe, multiple total internal reflections along the axis of a glass fibre of few microns radius in which index of refraction of core is greater than that of surroundings (cladding).



III. Mirage and optical looming :

Mirage is caused due to total internal reflection in deserts and other hot regions where, refractive index of air near the surface of earth becomes lesser than that above it due to heating of the earth. Light from distant objects approach the surface of earth with successively increasing i , till $i > \theta_c$ (Critical Angle) so that TIR takes place so that inverted images appear along with the objects as shown in figure.



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